

Better stock decisions for hotel bars

Krystal Ball assessment | 2023-01-01 to 2024-01-01 | 6 bars, 16 brands

1. Business problem and decision

Unavailable drinks damage service and sales; excess stock consumes cash and storage. This project estimates daily consumption by bar and brand, translates it into stock targets and compares replenishment outcomes. Prices and cost rates are unavailable, so no financial savings are claimed.

2. Data quality and assumptions

The source contains 6,575 rows and 96 bar-brand series over 366 days. 149 rows fail the balance equation at a 0.02 ml tolerance (maximum discrepancy 9.25 ml); consumption is retained and exceptions exported. There are 206 consecutive-record balance discontinuities.

81.3% of series-days have no log and are assumed zero consumption. Missing closing balances remain unknown. 416 observed zero-closing days with prior demand are stockout proxies, not confirmed lost sales. Recorded consumption understates latent demand during stockouts. Logging completeness is the principal deployment prerequisite.

3. Model selection and evidence

Chronological 60/20/20 train/validation/test split; weekly rolling origins predict up to seven days without future observations. Validation WAPE selects **mean_14**. Test starts 2023-10-20; it is not used to choose the winner. Forest inputs are past lags, rolling statistics, weekday and series identity. Holt-Winters uses fixed smoothing parameters.

Model	Test MAE (ml)	Test WAPE
mean_14	93.2	167.3%
seasonal_naive	97.5	175.1%
holt_winters	95.3	171.1%
random_forest	93.7	168.2%

WAPE is total absolute error divided by total consumption. Even predicting zero everywhere scores 100% WAPE here, beating all four candidates but providing no useful stock coverage. The chosen model is best only among the four candidates. Daily predictive skill is weak; validate logging and test weekly aggregation before a live pilot.

Inventory policy and rollout

4. Simulation performance and tradeoffs

Default par = next two days of forecast consumption + 1.645 x per-series validation RMSE x sqrt(2). Receive orders before demand; at day end replenish inventory position (on-hand plus on-order) to target. An end-of-day order arrives before demand two days later. Unmet demand is lost. Both policies start with identical seven-day-mean stock and no pipeline.

Test-period measure	Fixed 7-day target	Forecast + buffer
Stockout bar-brand-days	527	373
Lost volume (litres)	92.9	66.3
Volume fill rate	76.5%	83.2%
Mean closing ml / series	308.1	389.6
Period turnover	10.24	8.81

The forecast-and-buffer policy changes mean holding stock by +26.5% and stockout series-days from 527 to 373. The seven-day comparator is an assumed rule, not the hotel's measured current policy. Both replays use the same recorded consumption, so these outcomes do not prove causal savings or recover missing demand.

The nominal 95% normal factor is not a guaranteed fill rate. Sensitivity results cover 1-3 day lead times and factors 1.282, 1.645 and 2.326. Continuous-ml ordering is simulated; 750 ml bottle rounding is indicative only. No holding-cost optimisation, expiry, case packs, capacity limits or variable delivery times are included.

5. Production plan and failure modes

First reconcile inventory records and establish whether missing logs mean no demand. Pilot with manager review: daily ingestion, completeness checks, refreshed forecasts and targets, then actual stock and pipeline checks before ordering. Monitor forecast error, fill rate, stock, delivery delays and manual overrides. Freeze model changes until a chronological backtest passes agreed service and holding thresholds.

Next add occupancy, promotions and holidays; supplier calendars and lead-time variability; bottle sizes and order constraints; censored-demand estimation; and intermittent-demand alternatives. Correlated errors and supplier delays can make the square-root safety buffer too small. Product launches and closed bars need explicit availability calendars.

All 96 recommendations are historical as of 2024-01-01. They are not current purchase orders. Source: user-supplied Consumption Dataset - Dataset.csv; full methods, audit flags, sensitivity results and reproducible code accompany the notebook.